

HEMANTH THATHIREDDY

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PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 2+ years of experience designing scalable Java and Python services, distributed systems, and event-driven workflows on AWS. Experienced in building REST APIs, production-ready backend pipelines, and cloud-native services with a strong focus on performance, reliability, and operational excellence. Also, comfortable working across frontend integrations when needed, and known for improving system efficiency, debugging complex issues, and collaborating effectively with cross-functional teams.

TECHNICAL SKILLS

Frontend: React.js, Vite, JavaScript (ES2020+), TypeScript, HTML5, CSS3, Tailwind CSS, WebSockets, Responsive UI, Accessibility

Backend: Java, Spring Boot, Python, Go, REST APIs, OOP Design Patterns, Microservices, Authentication(JWT/OAuth2)

AWS & Cloud: Lambda, DynamoDB, Kinesis, EC2, S3, RDS, IAM, Event-Driven Architectures, Asynchronous Workflows, Serverless Architectures.

Databases: MySQL, PostgreSQL, DynamoDB, MongoDB, Redis, Hibernate/JPA, SQLAlchemy

Testing & DevOps: Jenkins CI/CD, Docker, Kubernetes, Git, SonarQube, Playwright, Vitest, JUnit, Mockito, Pytest

AI / LLM: OpenAI API Integration, NLP Pipelines, AI-Assisted Data Extraction, LLM-Powered Product Features

AI Development Tools: Claude Code, Cline, Kiro, Cursor

WORK EXPERIENCE

Software Development Engineer • Genx AI Solutions Corp • USA

Nov 2025 – Present

- Delivered AI-powered data processing features integrating OpenAI APIs and NLP services into **backend pipelines**, improving structured **data extraction accuracy by 35%** across large unstructured datasets.
- Created and implemented **Python REST APIs** with robust validation and fault handling, reducing manual analyst **processing time by 40%**.
- Guided technical design discussions for **scalable AI-assisted workflows**, influencing **architectural decisions** that improved maintainability and reduced **deployment iteration time by ~20%**.
- Collaborated with a **cross-functional team of 6+ engineers**, analysts, and stakeholders to deliver **iterative product enhancements** aligned with customer feedback.
- Leveraged **Claude Code** and **Kiro** to draft implementation ideas, speed up debugging, and improve backend development efficiency, reducing implementation and **troubleshooting time by ~25%**.

Software Engineer • Tek Labs Inc • USA

Mar 2025 – Oct 2025

- Led frontend development** of a React/Vite platform serving **10,000+ concurrent users**, improving page responsiveness by **30%** through optimized rendering, asynchronous data loading, and efficient state management.
- Architected a reusable UI component library** adopted across **4+ product modules**, reducing frontend feature delivery time by **40%** and improving consistency across customer-facing workflows.
- Established automated frontend testing pipelines** using **Vitest** and **Playwright**, reducing production UI regressions by **50%**.
- Engineered scalable **Java/Spring Boot REST APIs** and **backend services**, improving request **handling efficiency by ~20%** and supporting authentication, partner workflows, asynchronous processing, and third-party integrations.
- Built real-time **WebSocket event delivery** and **asynchronous backend workflows**, integrating **AWS Lambda** and **S3** to enable sub-second dashboard updates, **reduce polling load by 30%**, and improve processing efficiency by **25%**.

Full Stack Developer • Tata Consultancy Services • India

Feb 2022 – Dec 2022

- Developed React.js banking dashboards **supporting 1M+ daily financial transactions** with **real-time balance views**, portfolio analytics, and transaction reporting.
- Delivered **Java/Spring Boot microservices** for **payment settlement**, **SWIFT routing**, and account management in **distributed environments**, supporting high-volume financial transactions.
- Introduced **asynchronous event-driven transaction processing** pipelines using **Kafka/Kinesis-style streaming architectures**, reducing fraud **detection latency by 20%** and improving real-time processing reliability.

- Established customer feedback and reporting systems while applying **NLP-powered** document extraction workflows, improving business reporting workflows, operational visibility, and reducing manual **verification efforts by 30%**.
- Strengthened deployment stability and feature delivery by using **zero-downtime database** migration strategies and led Agile initiatives across **8+ engineers** through full production release cycles.

PROJECTS

Warehouse Management System — React, TypeScript, Java/Spring Boot, AWS

- **Designed and assembled a full-stack warehouse management platform** automating **invoicing**, inventory workflows, **route management**, shipment tracking, and **operational budgeting** for warehouse operations.
- Created scalable React dashboards with real-time operational visibility and backend services supporting **asynchronous workflow processing** and multi-user coordination.
- Developed **distributed Java/Spring Boot APIs** and **AWS-backed workflows** reducing operational effort by an estimated **45%** and improving invoice processing efficiency.
- Architected reusable frontend components and reporting modules, reducing feature **implementation time by ~30%** and improving **operational insights** for warehouse stakeholders.
- Built the Warehouse Management System using React, TypeScript, Java/Spring Boot, and AWS, and utilized **Google AI Studio** to assist with **workflow design, API development, and rapid iteration** on system features, reducing **manual operational effort by ~45%**.

EDUCATION

M.S. Computer Science • Illinois Institute of Technology, Chicago, IL

Jan 2023 – Dec 2024

Bachelor's in engineering • Siddharth Institute of Engineering and Technology